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PALAEOMAGNETISM OF MIDDLE PROTEROZOIC BASIC INTRUSIVES IN CENTRAL SWEDEN AND THE FENNOSCANDIAN APPARENT POLAR WANDER PATH

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ABSTRACT

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Palaeomagnetic results are presented from three sets of post-Svecokarelian intrusions in central Sweden. The first is a system of E—W trending doleritic dykes, including the large Breven and Hällefors dykes, and with a probable emplacement age of c. 1530 Ma. Their original NRM is disturbed by later events but two magnetic directions, Group EW1, with decl. 359.5°, incl. −6.2°, α_{95} = 8.2°, N = 35, and Group EW 2 with decl. 11.1°, incl. 51.3°, α_{95} = 10.1°, N = 14, have been isolated and give palaeopoles at 27.7° N, 196.8° E and 61.7° N, 176.1° E, respectively. The Group EW1 direction is considered to represent the original emplacement NRM, this is based on comparison with contemporary palaeomagnetic data from elsewhere in Fennoscandia.

The second set comprises dolerites and quartzporphyries, the Tuna dykes, age 1371 Ma. They present a stable magnetization with decl. 14.8°, incl. −18.1°, α_{95} = 9.6° and a palaeopole at 21.1° N, 180.4° E.

The third set studied are dolerites belonging to the Dala and Jämtland complexes of the Central Scandinavian Dolerite Group (CSDG), age c. 1250—1200 Ma (post-Jotnian). They present well-defined magnetic directions that give palaeopoles close to the equator around 155° E, in agreement with other published palaeomagnetic CSDG-data. However, there are deviating results obtained from dolerites belonging to the Dala complex of the CSDG. Those deviating have a mean decl. of 12.9°, incl. −1.4°, α_{95} = 18.1°, N = 5 which gives a palaeopole at 27.0° N, 179.5° E situated within the area of the 1600—1500 Ma poles.

Results from three, probably Sweconorwegian (c. 1000—850 Ma), doleritic dykes are also presented.

The results obtained are compared with other contemporary palaeomagnetic poles from Fennoscandia and a revised apparent polar wander path is presented and discussed.

INTRODUCTION

Several apparent polar wander paths (APWP) have been constructed for the Precambrian of Fennoscandia with the aim of obtaining information on the relative movements and orientations of Fennoscandia during the

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Proterozoic (see e.g., Poorter, 1981; Pesonen and Neuvonen, 1981; Piper, 1982). There are significant differences between the suggested APWPs and the variances have been attributed: (1) to problems in the interpretation

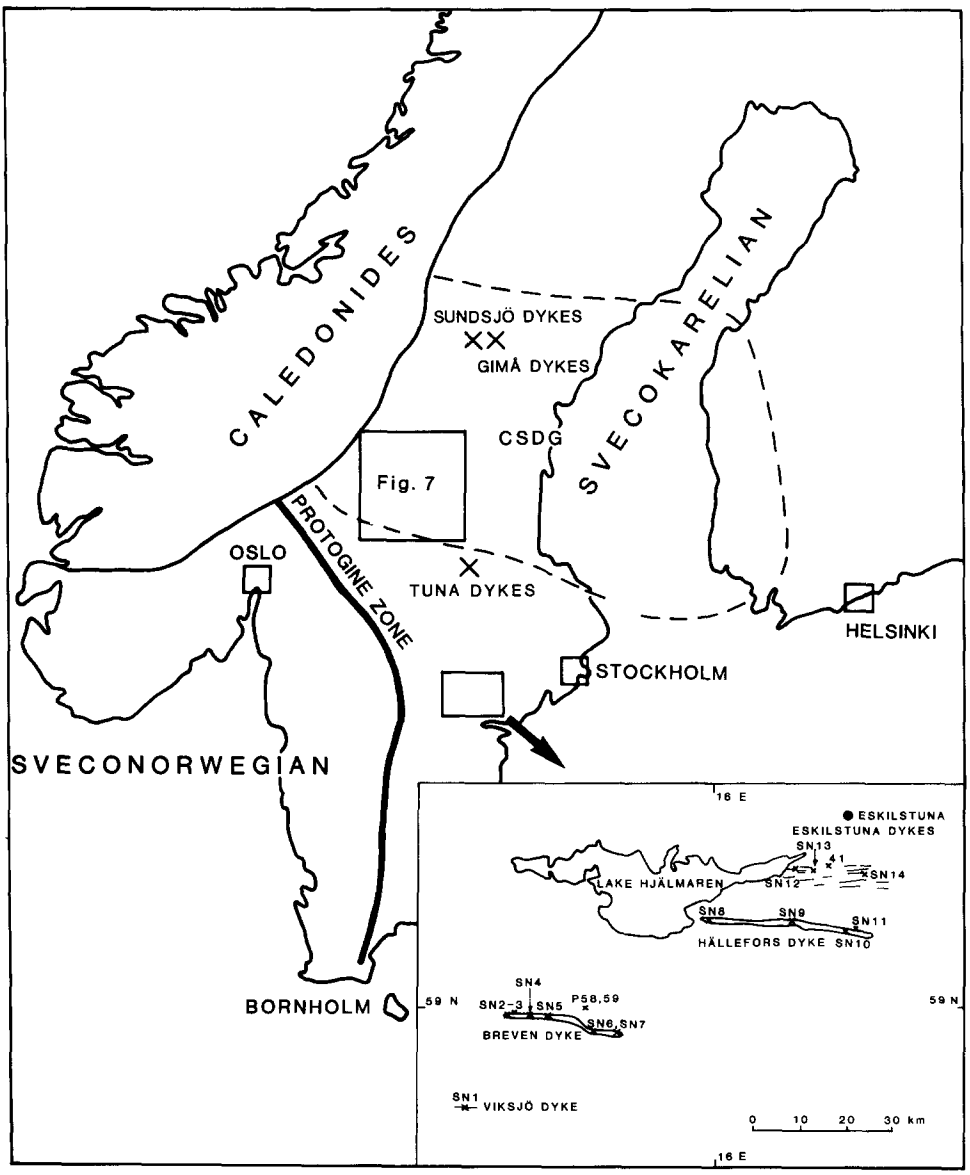


Fig. 1. General location map. The dashed line surrounds the approximative area for the CSDG-dolerites. Inset is the location map for the E—W trending dykes. Crosses denote sampled sites. P 58 and 59 are sites sampled by Piper (1980a, b) at Bo corresponding to sites 22 and 40 in Table II and Fig. 12. Site 41 is the Árby dolerite site presented by Patchett and Bylund, 1977.

of the radiometric ages presented for the rock units studied (e.g., Berger and York, 1980); (2) to lack of palaeomagnetic data for parts of the APWP; and (3) to difficulties in isolating relevant palaeomagnetic directions. In an attempt to improve a part of the Fennoscandian APWP, palaeomagnetic results from three sets of middle Proterozoic basic intrusions in central Sweden are presented and a revised APWP for the period c. 1650—1200 is discussed.

The rock units studied are all from areas little affected by tectonic and magmatic events since the end of the Svecokarelian orogeny at c. 1700 Ma. There are: (1) a system of E—W trending dykes situated c. 150—200 km W of Stockholm (Fig. 1); (2) a locally developed group of NNE-trending dykes, the Tuna dykes, in the province of Dalarna 200 km NW of Stockholm (Fig. 1); and (3) dykes and sills belonging to the post-Jotnian Central Scandinavian Dolerite Group (CSDG) in the provinces Dalarna and Jämtland (Figs. 1 and 7). Also included in the study are results from two dykes that presented Sveconorwegian palaeomagnetic directions and a dyke of Sveconorwegian radiometric age but with an anomalous magnetic direction.

A comprehensive description of the geology of central Sweden is given by Lundqvist (1979).

SAMPLING AND MEASUREMENT

Sampling was done either with a handheld coredrill or by hammering the sample loose. The *in situ* orientation was done with a magnetic compass or with a suncompass. Most sampling was done in roadcuts or quarries presenting fresh outcrops. From each core or handsample one to four specimens were obtained.

Measurements were made either with a 85 Hz UGF-JR3 or a 7 Hz Digico spinner magnetometer. Progressive alternating field (AF) or thermal cleaning was used for isolating a stable characteristic remanent magnetization. Peak alternating fields up to 100 mT (millitesla) or heating up to 670°C were used. In addition, a number of samples were first AF-demagnetized to 20—50 mT and then thermally demagnetized. During the AF-treatment the specimens were tumbled in a two-axis tumbler. All demagnetization was made in fields < 10 nT.

Fisher's (1953) statistics were used in calculating mean magnetic directions. All magnetic data in Table I are based on sample means combined to site means.

An estimate of the Curie temperature was made by measuring the saturation magnetization (J_S) against temperature. A saturation field of 0.8 T was used and the measurements were made in air.

TABLE I

Magnetic directions before and after demagnetization

Site	N	AF (mT)	Decl. (°)	Incl. (°)	α_{95} (°)	k	R
The E—W trending dykes							
The Viksjö dyke							
SN1, Mårsätter	5	0	95.5	60.2	14.3	29.5	4.86438
	5	50	94.0	59.4	14.2	30.1	4.86703
The Breven dyke							
SN2, Rävsjön	6	0	323.7	71.4	21.1	11.0	5.54705
	6	30	348.5	1.6	13.9	24.2	5.79321
SN3, Korttorp	7	0	353.4	−3.8	22.1	8.4	6.76390
(1548 ± 23, 1542 ± 23)	7	50	335.5	−37.9	7.4	67.6	6.91122
SN4, Sjölunda	8	0	347.8	38.4	10.6	28.3	7.75235
	8	60	350.7	20.6	6.4	74.8	7.90646
SN5, Bottorp	7	0	359.4	34.9	65.2	1.8	3.69285
	7	20	5.6	13.0	24.4	7.1	6.15065
SN6, L. Kringsjötorp	2	0	10.0	76.3	—	—	—
	2	20	33.8	65.9	—	—	—
SN7, Eriksberg	2	Unstable					
The Hällefors dyke							
SN8, Gluggen	4	0	40.8	65.2	16.2	33.1	3.90947
	4	10	50.5	62.4	19.6	23.0	3.86982
SN9, Tyckenhed quarry	4	0	354.6	33.2	27.9	11.8	3.74584
	4	20	352.9	−4.9	7.3	160.8	3.98134
SN10, Hälleforsnäs	14	0	338.7	53.7	41.1	1.9	7.18395
(1518 ± 38)	14	50	354.3	37.8	32.1	2.5	8.77664
SN11, Ölmestorp	2	0	37.1	13.1	—	—	—
	2	40	32.0	4.3	—	—	—
The Eskilstuna dykes							
SN12, Gåstorp	1	0	35.0	46.7	—	—	—
(1532 ± 22)	1	40	359.6	12.6	—	—	—
SN13, Bälgviken	1	0	36.7	33.3	—	—	—
	1	40	25.0	−13.0	—	—	—
SN14, Forskilen	7	0	48.3	58.2	9.2	44.4	6.86480
(1560 ± 79, 1510 ± 23)	7	30	15.7	14.8	11.2	29.8	6.79875
Samples stable to 100 mT and 500—700°C. Grouped according to inclination							
Group EW1, shallow inclination							
AF-demag.	23	100	356.0	−4.9	9.7	10.7	20.93932
Thermal demag. 500—570°C	12		6.3	−8.6	15.7	8.6	10.72369
Mean AF and thermal demag.	35	—	359.5	−6.2	8.2	9.8	31.53801
Group EW2, intermediate inclination							
AF-demag.	14	100	4.7	53.8	11.1	13.8	13.05522
Thermal demag. 500—570°C	4		28.3	40.4	25.4	14.1	3.78709
Mean AF and thermal demag.	18	—	11.1	51.3	10.1	12.6	16.65058

Site	N	AF (mT)	Decl. (°)	Incl. (°)	α_{95} (°)	k	R
TD, Tuna dykes	9	0	27.2	41.1	18.5	8.7	8.08102
(1371 ± 50)	9	30	14.8	−18.1	9.6	29.9	8.73258
CSDG dolerites, Dala complex							
D1, Emådalen	7	0	324.4	55.1	56.9	2.1	4.11764
(1223 ± 36)	7	30	2.3	−2.7	15.7	15.7	6.61681
D2, Älvho	5	0	32.2	−19.2	22.6	12.4	4.67855
(1290 ± 63, 1215 ± 18)	5	50	23.6	−16.4	12.1	41.1	4.90267
D3, Älvdalsåsen	11	0	50.7	−27.1	13.0	13.3	10.24969
(1231 ± 18)	11	30	49.5	−37.6	5.3	74.0	10.86493
D4, Mossi	5	0	55.9	13.0	98.5	1.6	2.46397
(1217 ± 18)	5	40	45.3	−24.3	35.1	5.7	4.29178
D5, Bunkris	8	0	356.5	2.2	3.5	245.1	7.97145
(1516 ± 62; 1546 ± 84)	10	10	1.5	−8.7	3.3	218.4	9.95880
D6, Glysjön	5	0	39.6	23.1	24.0	11.2	4.64157
	6	50	11.1	16.9	8.2	68.4	5.92690
Magnetic directions for CSDG dolerites, the Dala Complex presented by Mulder (1971) and discussed in the text							
ZDA, Ämån	5	—	28	4	7	138	—
ZDC, Bunkris (= D5)	4	—	340	−5	17	38	—
ZDD Lybergsgnupen	5	—	31	−42	10	62	—
ZDE Glysjön (= D6)	5	—	21	15	6	148	—
Mean of Dala complex directions deviating from the established CSDG-pattern Includes sites D1, D2, D5, D6 and ZDA							
	5	—	12.9	−1.4	18.1	18.7	—
CSDG dolerites, Jämtland complex							
J1, Sundsjö	5	0	47.2	−31.6	74.4	2.0	3.01926
(1156 ± 15, 1167 ± 19, 1213 ± 18)	5	50	40.2	−52.2	16.7	21.9	4.81756
J2, Gimån	18	0	53.1	4.8	19.0	4.3	14.04485
(1229 ± 19, 1179 ± 56)	18	40	51.4	−41.2	8.1	19.4	17.12339
Sveconorwegian dolerites							
D7, Nornäs	4	0	330.1	73.0	5.7	259.1	3.98842
(901 ± 39, 933 ± 13)	4	10	325.0	71.3	6.5	203.4	3.98525
Piper and Smith, 1980 site 43 = D7	6	—	0.4	82.2	3.1	459	5.99
D9, Ejen	8	0	109.2	41.1	39.5	2.9	5.60787
	8	20	127.6	23.7	8.5	43.4	7.83882

N = number of samples. AF = demagnetizing field in milliteslas with least scatter in directions and presented together with data obtained before demagnetization. α_{95} = semi-angle of the 95% cone of confidence. k = the best estimate of Fisher's (1953) precision parameter. R = length of the resultant vector, each sample given unit weight. The ages presented together with sites are Rb—Sr whole rock and mineral ages given by Patchett (1978).

THE E—W TRENDING DYKES

The E—W trending dykes occur in an area 120 km long (E—W) and 60 km wide (N—S) in the provinces Södermanland and Närke, WSW of Stockholm (Fig. 1). The dyke complex consists of two large dykes, the Breven and Hällefors dykes and various smaller dykes, the Eskilstuna dolerites. Country rocks are Svecokarelian migmatites and granites.

The Breven dyke, which is c. 27 km in length and up to 1 km wide, is composed of a western and a central part, made up entirely of olivine dolerite, and an eastern part with a central granophyre body surrounded by olivine dolerite. In addition a variety of hybrid rocks have been produced by mixing of the granophyre with the dolerite (Krokström, 1932).

The Hällefors dyke is about 40 km in length and 1 km wide. It consists of a marginal olivine dolerite and a central part porphyric in feldspar and pyroxene (Krokström, 1936).

The Eskilstuna dolerites vary between a few cm to 100 m in width and include dolerites both with and without olivine and plagioclase-phyric dolerites also with or without olivine (Gorbatschev, 1961).

The E—W trending dykes have been dated by Patchett (1978) by the Rb—Sr whole rock and mineral methods, and he gives c. 1530 Ma as a probable emplacement age for the dyke system.

Piper (1980a) presented palaeomagnetic data from a section across the eastern part of the Hällefors dyke (Fig. 1).

Palaeomagnetic results

The palaeomagnetic results are listed in Table I and the behaviour of typical specimens during progressive demagnetization are plotted in Figs. 2 and 3.

The general trend during AF-demagnetization is that a soft component is removed in low peak fields of the order of 5—10 mT (Fig. 3a). This is accompanied by a change from a steep to intermediate positive inclination to a shallow positive or shallow or intermediate negative inclination (Fig. 2a, sample SN3:103 and SN10:143 and Fig. 3, sample SN10:3). See also Table I where initial directions of the natural remanent magnetization (NRM) are listed together with magnetic directions from the levels with the least scatter, as given by the parameters α_{95} and k. The same directional trend is furthermore observed in the thermal demagnetization, where the steep component is more resistant to heating than to AF-treatment but removed at 400—500°C, leaving a weak shallow component (Fig. 2b, sample SN5:123 and Fig. 3, sample SN2:31). There is, however, a large number of samples that retain their steep original direction during both AF and thermal stepwise demagnetization, e.g., sample SN5:121 in Fig. 2 and sample SN8:11 in Fig. 3.

In the western part of the Breven dyke, sites SN 2—5, Fig. 1, the directional trend is from an intermediate positive inclination to shallow or intermediate negative inclination. This is the olivine dolerite part of the dyke which is separated from the eastern granophyric part by a fault (Krokström, 1932). The results from this latter part, sites SN6—7, are obscure and several samples, including all from site SN 7, had to be discarded due to magnetic instability. Two granophyre samples from site SN 6 presented steep to intermediate positive stable directions (Fig. 3, sample SN6:123).

The Hällefors dyke, sampled at four sites, displayed a nonuniform pattern. The dolerite at the westernmost site, SN8, gave a steep initial positive inclination that stays steep during demagnetization but with an increased scatter. The other sites present the typical pattern with a steep to intermediate positive inclination changing to a shallow inclination. This pattern is repeated in the Eskilstuna dykes, sites SN 12—14.

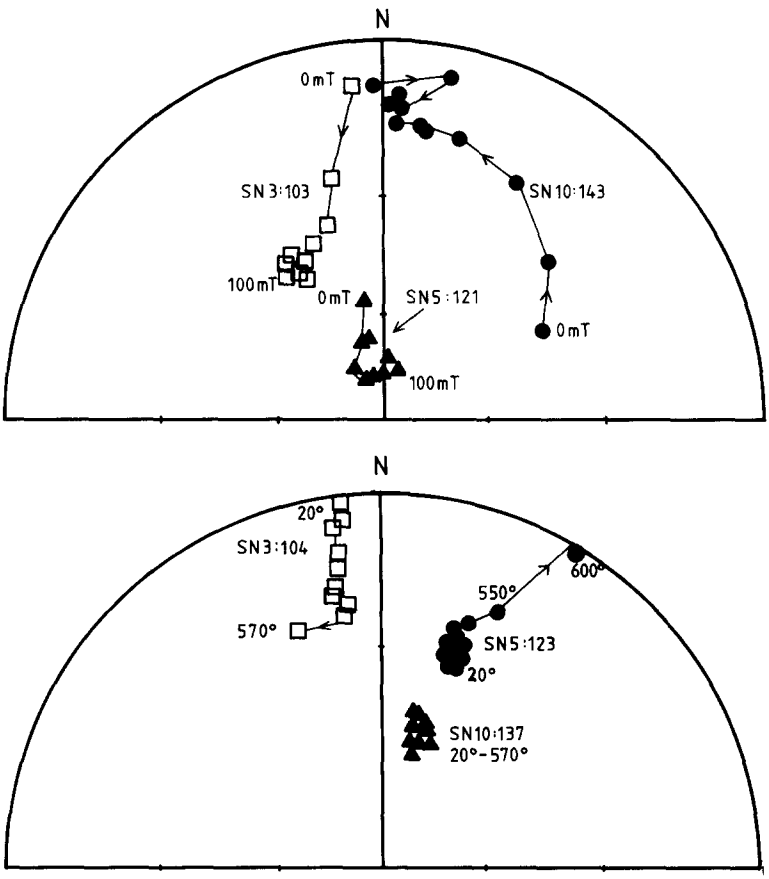


Fig. 2. Behaviour of pilot specimens from the E—W dykes during progressive alternating field (a) and thermal (b) demagnetization. Closed symbols denote positive inclination, open symbols negative inclination. Stereographic projection.

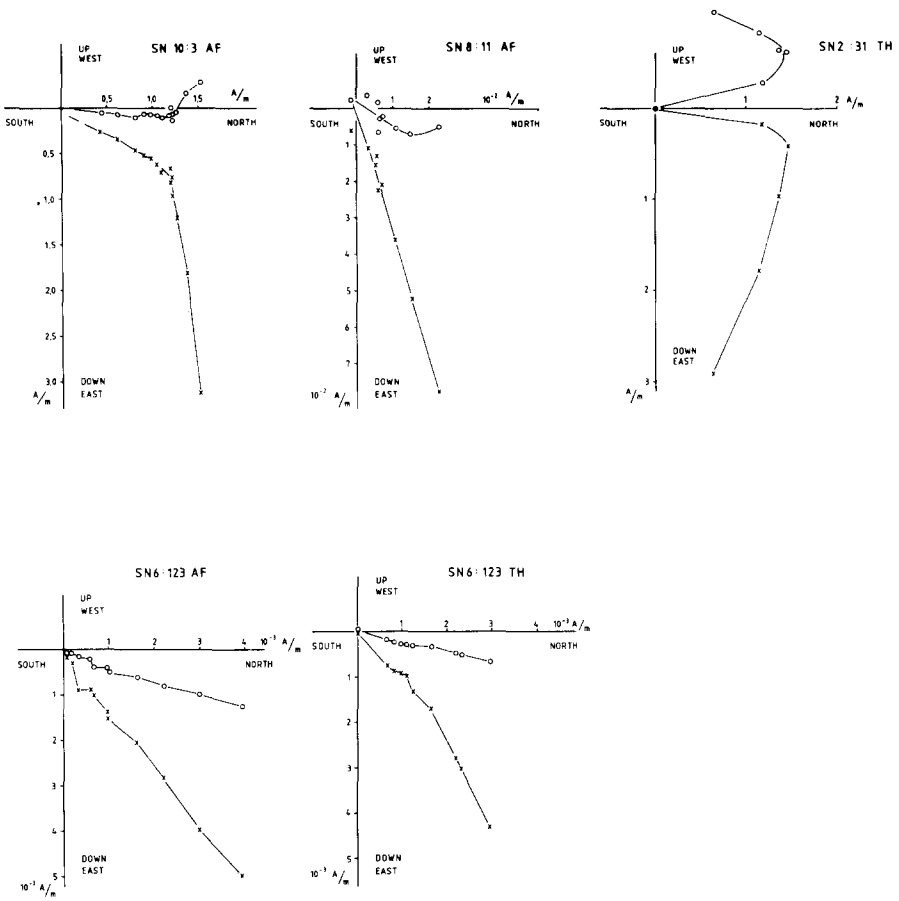


Fig. 3. Orthogonal projections (Zijderveld diagrams) of the end points of the natural remanent magnetization vectors found in the E—W dykes during progressive demagnetization. Crosses plot in the vertical N—S plane, circles in the horizontal W—N plane.

The data from the E—W dykes presented in Table I show, at least partly, a large scatter after demagnetization, e.g., sites SN5, 8 and 10. This may be due to an anhysteretic remanent magnetization picked up during the laboratory procedures or reflect a complex NRM difficult to resolve into characteristic magnetic components with the methods used. There are, however, a substantial number of samples from all dykes that were magnetically stable and presented significant endpoints in fields up to 100 mT or temperatures between 500—570°C. Those samples can be divided into two groups, Group EW 1 and Group EW 2, with the first being composed of those that during demagnetization obtained stable directions with shallow positive or negative inclination pointing north, while Group EW 2 comprises those which were more or less stationary during demagnetization and ended up with a direction of intermediate inclination pointing NNE (Fig. 4).

Also listed in Table I is a soft component obtained by vector subtraction from specimens of Group EW 1 during the initial steps of AF-demagnetization. This component may be due to influences from the Earth's present field or to fields picked up during storage in the laboratory.

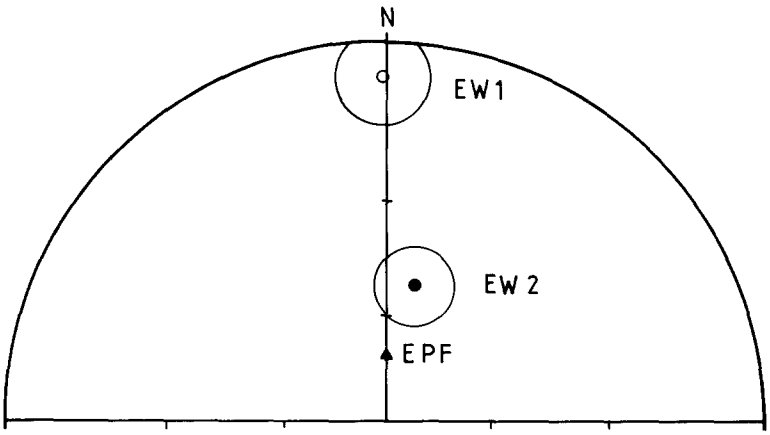


Fig. 4. The two groups of magnetic directions found in the E—W dykes after AF- and thermal demagnetization to 100 mT or 500—570°C. Circles are α_{95} confidence circles. EPF = direction of the Earth's present field. Conventions as in Fig. 2.

The thermal demagnetization curves display two magnetic phases, one with a Curie temperature at c. 250°C and another around 570°C, the Curie temperature for nearly pure magnetite. Ore microscopy reveals magnetite and ilmenite both as separate grains and as exsolution lamellae. Further martite and haematite has been observed. Of the sulphides, pyrite and pyrrhotite has been found.

The large decrease in intensity during the first steps in the AF-demagnetization may be due to multidomain magnetite grains with low coercivity while the remaining hard magnetization may be ascribed to single domain and pseudo-single-domain magnetite. Haematite may also contribute to the hard magnetization component observed. In his study of the Hällefors dyke, Piper (1980a) proved the presence of haematite as a high coercive and high temperature component of the characteristic magnetization of this dyke. The direction of this haematite component was approximately decl. = 300° and incl. = −32°.

Saturation magnetization (J_S) versus temperature curves show a flexure at c. 250°C, indicating a low temperature component.

The study of the magnetic minerals does not present any reason for the division into two groups of different magnetic directions. Furthermore both directions can be found in samples from the same site.

Patchett (1978) discussed possible thermal and hydrothermal disturbances that can have affected the isotopic system in the Breven and Häl-

lefors dykes after their displacement. Such events may likewise have affected the original NRM of the dykes both thermally and chemically and induced a complex magnetism in the rocks.

THE TUNA DYKES

The Tuna dolerite and quartz-porphyry dykes occur as a well-defined dyke swarm in the southern part of the province Dalarna (Fig. 1). They are from a few cm to 15 m wide and trend generally NNE. The swarm consists of porphyritic dolerites often associated with quartz-porphyries, the Gustafs porphyries, and sometimes they form composite dykes. The Tuna dykes are dated to 1371 ± 50 with the Rb—Sr whole rock method (Patchett, 1978). Country rocks are Svecokarelian granites and leptites (Hjelmqvist, 1950, 1966).

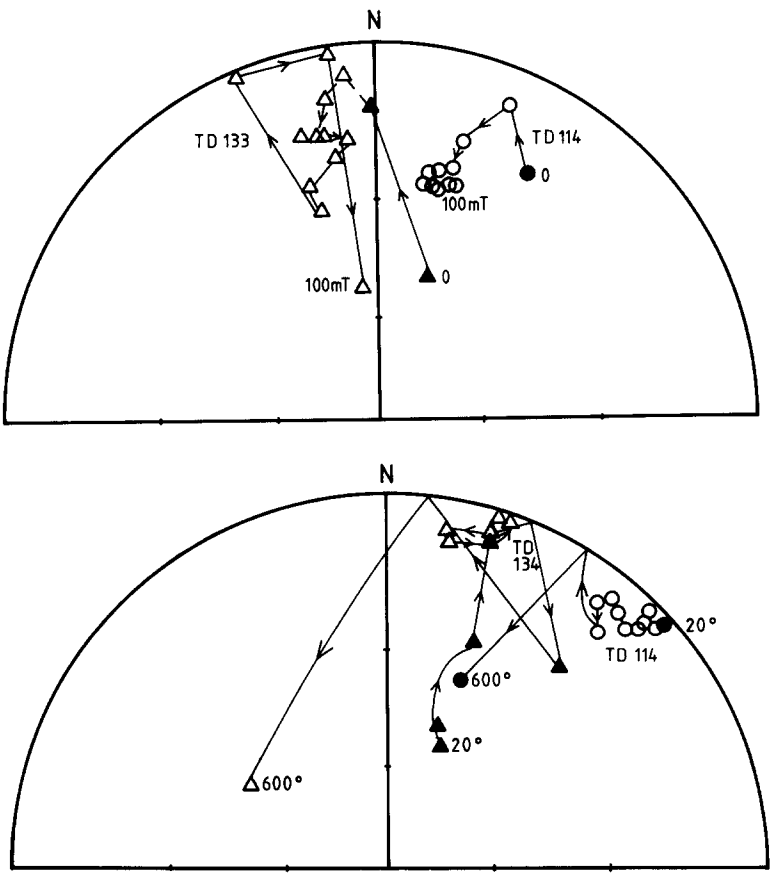


Fig. 5. Behaviour of pilot specimens from the Tuna dyke dolerites (TD 114) and quartz-porphyries (TD 133 and 134) during progressive alternating field (a) and thermal (b) demagnetization. Conventions as in Fig. 2.

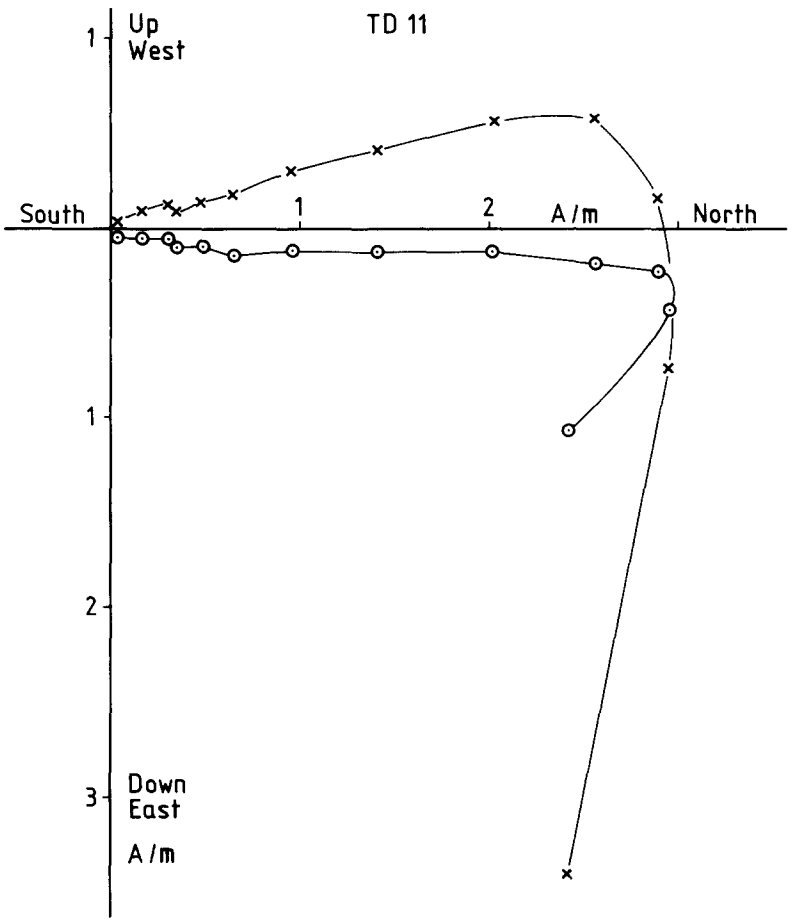


Fig. 6. Orthogonal projections of the end points of the natural remanent magnetization in a Tuna dolerite dyke during progressive AF-demagnetization. Conventions as in Fig. 3.

Palaeomagnetic results

The palaeomagnetic data are listed in Table I and plotted in Figs. 5 and 6.

The Tuna dykes contain a large soft magnetic component easily removed during the first steps of AF-demagnetization leaving a stable characteristic magnetization with shallow negative inclination pointing N—NNE (Figs. 5a and 6). This direction is found in both the dolerites and the quartz-porphyries. The thermal demagnetization gives the same directional trend (Fig. 5b). It also reveals the presence of magnetic phases with low blocking temperature in both the dolerite and the quartz-porphyry, a matter that is not clearly observed in the J_s—T measurements. The microscopic study shows that ilmenite and titanomagnetite are the main opaques in the dolerite

while the opaque grains in the quartz-porphyry were too small to identify. The J_S—T measurements of the quartz-porphyry indicates the presence of a phase with blocking temperature > 600°C, probably haematite.

CENTRAL SCANDINAVIAN DOLERITE GROUP (CSDG)

The CSDG comprises an extensive group of alkali-basaltic olivine dolerite sills and dykes in central Sweden and western Finland. They form the

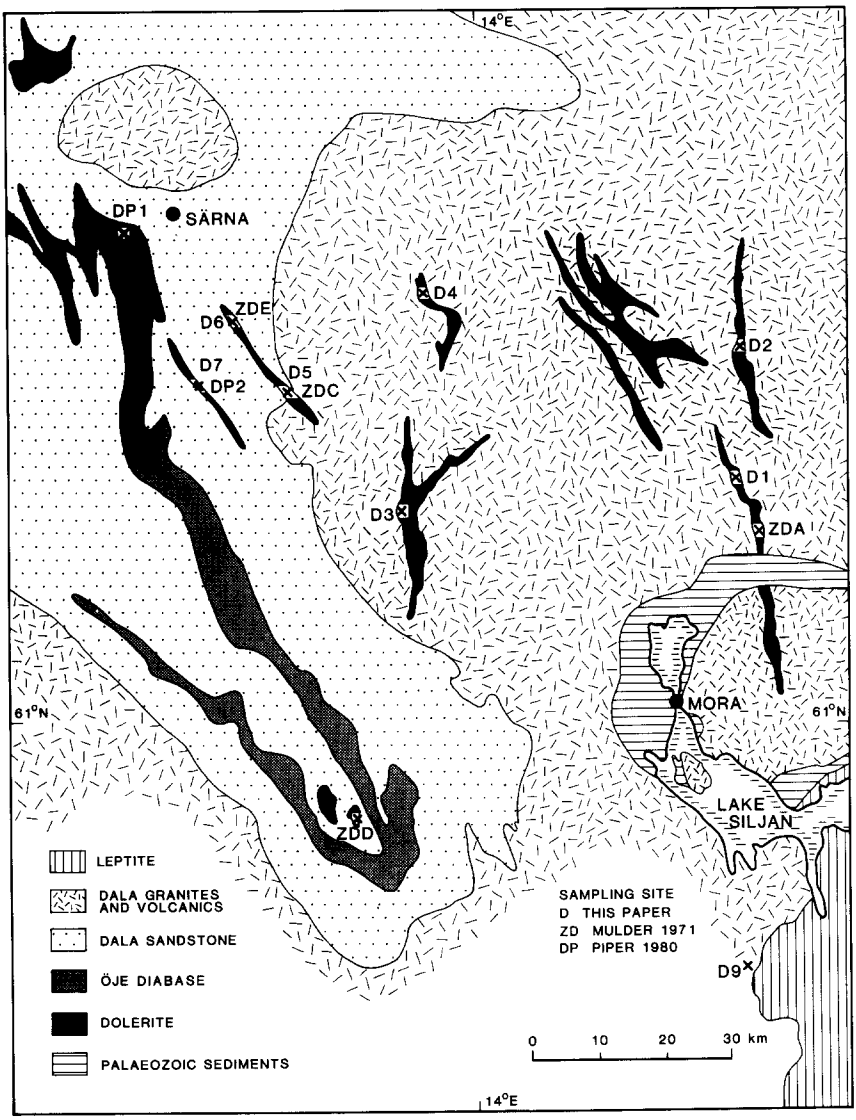


Fig. 7. Geological map of the area of the Dala Complex of the CSDG-dolerites with sampling sites.

Scandinavian part of a large magmatic event around 1270—1200 Ma in the Northern Hemisphere, manifested by the intrusion of extensive swarms of mafic sills and dykes (Patchett et al., 1978). The Scandinavian part

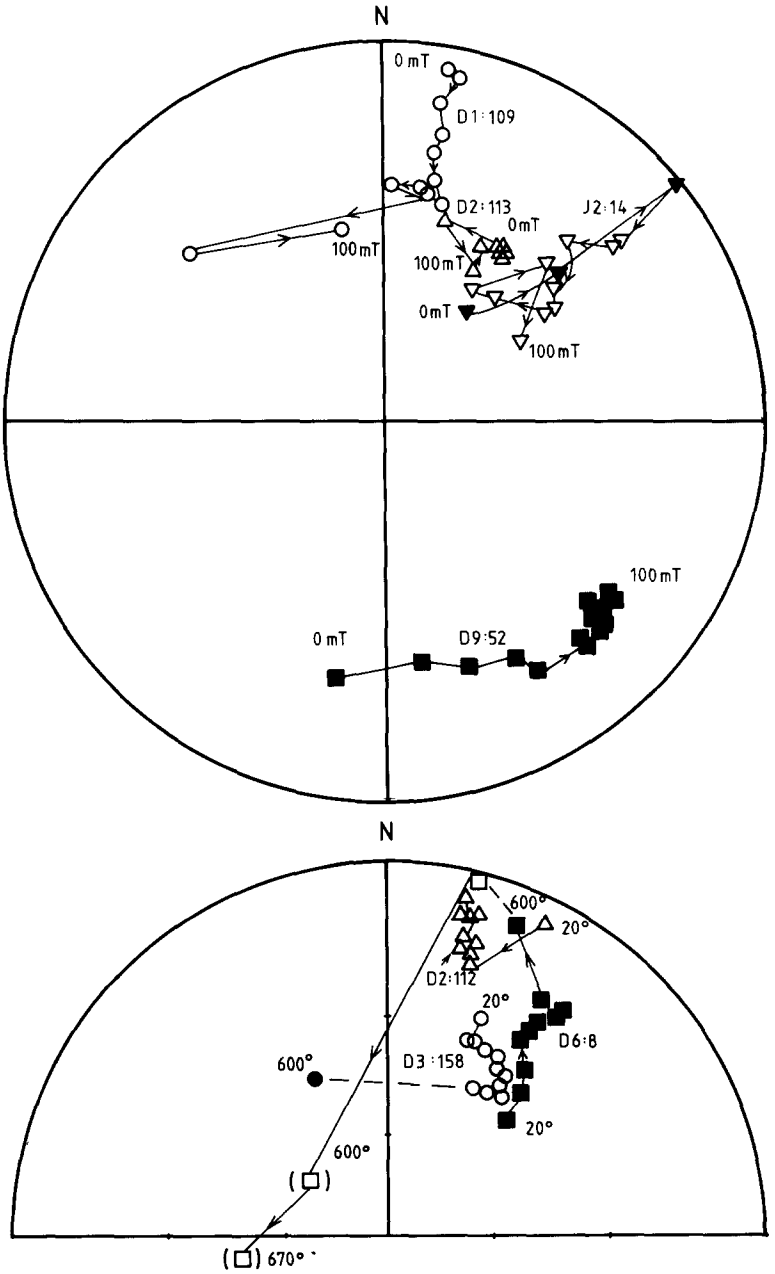


Fig. 8. Behaviour of pilot specimens from the CSDG-dolerites during progressive alternating field (a) and thermal (b) demagnetization. Conventions as in Fig. 2.

has been divided by Gorbatschev et al. (1979) into the Dala, Jämtland and Ulvö complexes in Sweden and the Satakunta complex in Finland. The terms Jotnian and post-Jotnian have also been used for these units (e.g., Mulder, 1971; Piper, 1979a; Piper and Smith, 1980). In this paper data are presented from the Dala and Jämtland complexes. In Dalarna (the Dala complex) the dolerites intruded Subjotnian lavas, age 1634 Ma (Welin and Lundqvist, 1970), Subjotnian granites, and Jotnian sandstones (Fig. 7). In Jämtland they intruded late Svecokarelian granites, e.g., the Revsund granites, dated at 1747 ± 40 Ma with the Rb—Sr method (Welin et al., 1971).

Palaeomagnetic results

The palaeomagnetic results are listed in Table I and plotted in Figs. 8—11.

The CSDG dolerites presented a simple magnetic pattern during demag-

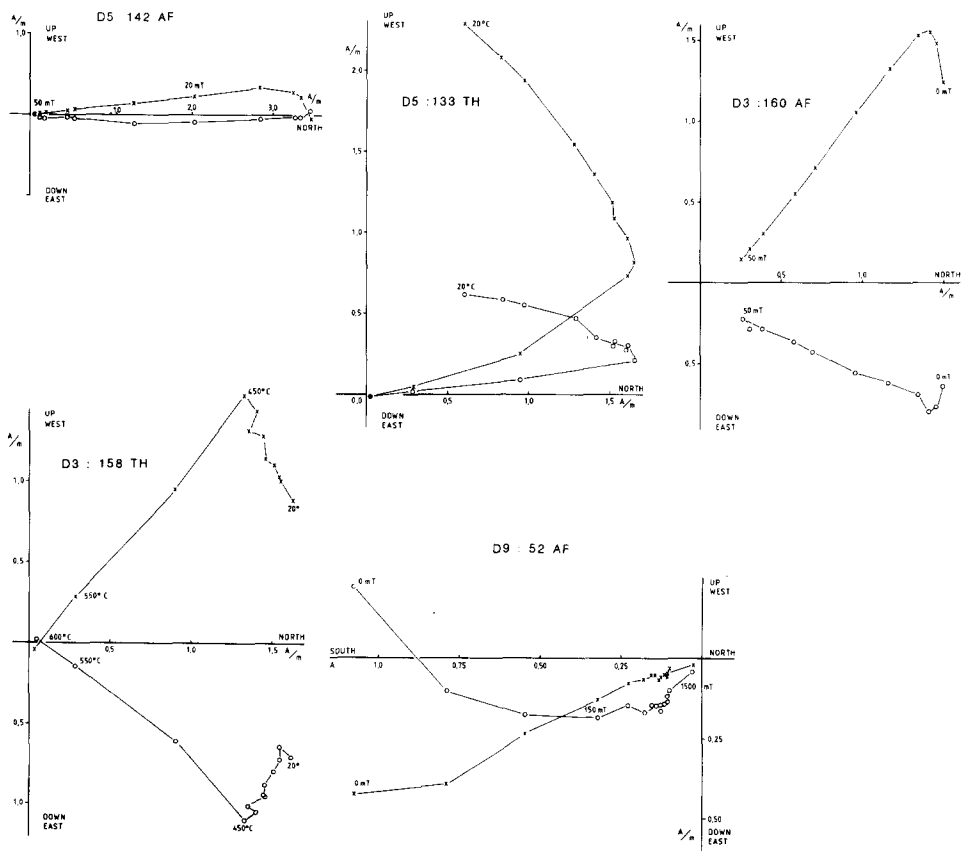


Fig. 9. Orthogonal projections of the end points of the natural remanent demagnetization vectors found in the CSDG-dolerites during progressive demagnetization. Conventions as in Fig. 3.

netization. A soft component was removed in low AF-fields isolating a stable characteristic magnetization (Figs. 8a and 9). The thermal demagnetization presented the same pattern (Figs. 8b and 9) and the demagnetization data together with the J_s—T measurements and ore microscopy indicate titanomagnetite as the main carrier of the magnetic remanence. A porphyritic dyke at site D2 and a rheomorphic granitic dyke in the dolerite at site J1 gave the same magnetic directions as the surrounding dolerite (Fig. 10).

There is, however, a distinct grouping in magnetic directions in the dolerites studied with one group, consisting of sites D3, D4, J1 and J2, all

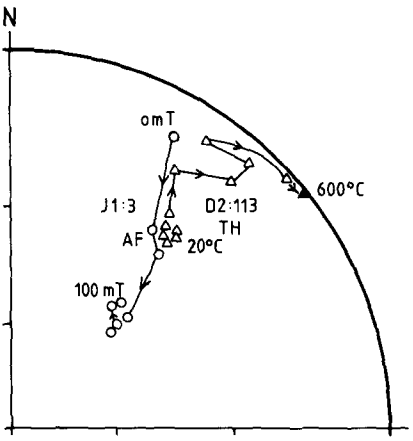


Fig. 10. Changes in direction during progressive demagnetization for a rheomorphic porphyritic dyke (D 2:113) and a granitic dyke (J 1:3) in two CSDG dolerites. Conventions as in Fig. 2.

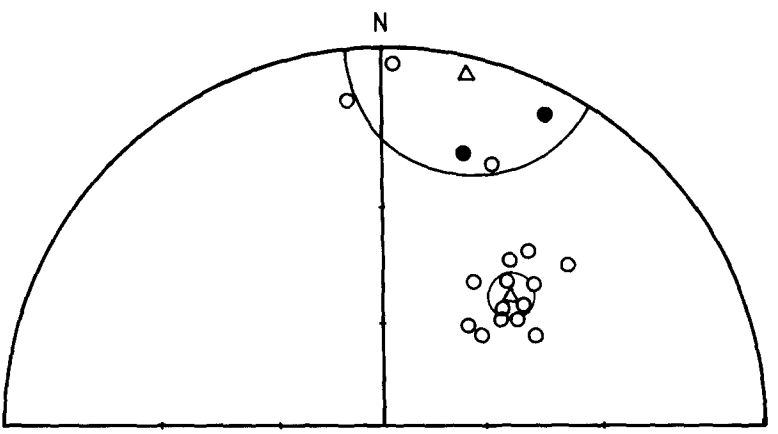


Fig. 11. The two magnetic directions found in the CSDG-dolerites. A is the main CSDG direction and B the deviating direction found in some of the dolerites of the Dala Complex. Triangles denote mean directions. Conventions as in Fig. 2.

with a negative intermediate inclination pointing northeast while sites D1, D2, D5 and D6 have a shallow inclination pointing north (Fig. 11). All sites in the latter group belong to the Dala complex.

SVECONORWEGIAN DOLERITES

During the field work, samples were collected from two dolerite dykes, sites SN1 (Mårsätter, Viksjö dyke; Fig. 1) and D9 (Ejen dyke; Fig. 7), that after demagnetization gave Sveconorwegian magnetic directions (Figs. 8b and 9, sample D9:52, and Table I). Further, samples were collected from a dolerite in the province of Dalarna (site D7; Fig. 7) that was dated by Patchett (1978) to 901 ± 39 Ma. This dyke had previously been sampled by Piper and Smith (1980) and is presented as their site No. 43. In both studies from that dolerite a magnetic direction with steep positive inclination was obtained.

THE APPARENT POLAR WANDER PATH; A DISCUSSION

The palaeomagnetic data from the studied middle Proterozoic intrusions in central Sweden help to clarify previously published APWPs for Fennoscandia (e.g., Piper, 1980a; Pesonen and Neuvonen, 1981).

The obtained palaeopoles are listed in Table II and plotted in Fig. 12 together with available palaeopoles from other rock units in Fennoscandia covering the same time range. Numbers in parentheses refer to pole numbers listed in Table II.

The APWP in Fig. 12 begins in an area with consistent palaeomagnetic data from mafic rocks, age c. 1900—1800 Ma, in northern Sweden and Finland (Piper, 1980a; Pesonen and Neuvonen, 1981; Elming, 1982). The path continues with the Kuisaari dolerite (3) and the Åland rapakivi granite (4) poles with ages of c. 1650—1550 Ma, and the Kumlinge poles (5 and 6), age c. 1600—1550 Ma.

After the Finnish 1650—1550 Ma poles there follows a prominent loop based on data from a set of rapakivi granite intrusions with associated anorthosites and gabbros from central Sweden, with the turnover of the loop founded on results from the Nordingrå intrusives (13 and 14). Preliminary Rb—Sr isochron ages of 1552 ± 50 Ma and 1414 ± 20 Ma have been reported for, respectively, the Nordingrå anorthosite-gabbro and the Nordingrå granite (Kornfält, 1976). Lundqvist (1980) gave a preliminary age of c. 1550 Ma for the Nordingrå granite, this age is based on complimentary U—Pb determinations on zircon. The Ragunda intrusion (15), situated c. 110 km west of the Nordingrå district, is of the same type as the Nordingrå intrusives and a contemporaneous emplacement time is possible, although Kornfält (1976) reported a Rb—Sr isochron age of 1292 ± 30 Ma for the Ragunda rocks. The Gävle granite (16) is a rapakivi granite similar to those at Ragunda and Nordingrå. The Gävle granite is not dated. It is questionable,

however, if the NRM measured in the Nordingrå, Ragunda and Gävle massifs, which form the loop, reflects the ambient geomagnetic field at the time of the intrusion of the massifs. It may represent a later blocking in of a NRM due to a later episode of uplift and cooling of the massifs. A tectonic rotation of e.g., the Nordingrå block is possible, but such a rotation, as also an episode of cooling and uplift, must have occurred before the intrusion of the dolerites of the Ulvö complex of the CSDG. Those dolerites cover parts of the eastern section of the Nordingrå anorthosite-gabbro and present pole positions (28—31) that agree well with the majority of the CSDG-poles. Further, the eastern part of the Nordingrå anorthosite-gabbro has been remagnetized by the capping dolerite, giving the anorthosite-gabbro in this area the same magnetic directions as the CSDG-dolerites (Piper, 1979a; Magnusson, 1983). No geological evidence has been presented for a major rotation or tilting of the Nordingrå block. The Jotnian Nordingrå sandstone that partly caps the anorthosite-gabbro dips about 15° SE. Correcting for this tilt does not significantly change the APWP-loop. A question is the relation between the Swedish anorthosite-gabbros and rapakivi granites and their Finnish counterparts. The latter are supposed to have intruded between c. 1700—1540 Ma (Vaasjoki, 1977). The Åland rapakivi massif has an U—Pb age of 1670 ± 20 Ma (Vaasjoki, 1977), which is in agreement with its palaeoposition (pole 4, Fig. 12). Further, Vaasjoki presented U—Pb ages of 1570 and 1540 Ma for the Laitala and Ytö rapakivi granites in western Finland. No palaeomagnetic data are at present available for those intrusives. Thus the age data on the Finnish rapakivi granites do not contradict an age of c. 1550 Ma for their Swedish counterparts as suggested by the preliminary datings presented above.

Regarding the loop it is the author's opinion that it reflects an apparent polar wander relative to Fennoscandia some time during the period c. 1650 Ma, as given by the Finnish age data and c. 1250 Ma, the time for the intrusion of the CSDG. This loop was first presented by Pesonen (1978) as the Sub-Jotnian loop. The clockwise direction of the path presented in Fig. 12 is based on the assumption that the magnetization in the Swedish rapakivi granites and associated rocks is primary and older than 1540 Ma. However, an anti-clockwise turn is possible, assuming the magnetization to be younger due to a later sequence of uplift and cooling.

The APWP may then continue northwards from the pole of the Ragunda intrusives (15) and over the poles of the undated Bo diorite (22) and the Group EW2 pole as suggested by the dashed curve segment in Fig. 12. The Bo diorite is situated in the vicinity of the Breven dyke (Fig. 1), which supplies a considerable part of the data for the Group EW2 pole, and this closeness both in geographical and palaeopole position suggests a similar magnetic history for the two rock units.

The next pole (25) in the dashed "hump" is from the dolerites that intrude the Ragunda anorthosite-gabbro. They give two magnetic directions, the other represented by pole 24. This led Piper (1979b) to consider at least

TABLE II

Fennoscandian palaeomagnetic poles used for comparison and for construction of the APWP in Fig. 12

Pole No.	Rock unit	Palaeomagnetic pole		d_p (°)	d_m (°)	Age (Ma)	Reference palaeomagnetic data
		lat. (°)	long (°E)				
1	Dundret gabbro, S	21.1 N	187.2	3.9	7.7	1530	Elming, 1982
2	St. Lulevattnet gabbro, S	23.0 N	200.0	4.0	8.0	1530	Elming, 1982
3	Kuisaari dolerite, F	22.0 N	190.0	7.0	14.0	1650	Neuvonen, 1978
4	Åland rapakivi, F	19.5 N	194.4	15.0	18.0	1659—1589	Johnson, 1979
5	Kumlinge-Brändö dikes, F	12.2 N	182.0	5.0	9.0	1602—1556	Pesonen and Neuvonen, 1981
6	Kumlinge dikes, F	14.3 N	190.1	8.0	14.0	1602—1556	Neuvonen, 1970
7	Föglö-Sottunga dikes, F	27.9 N	187.5	6.0	13.0	1523—1518	Pesonen and Neuvonen, 1981
8	Föglö dikes, F	31.3 N	186.5	6.0	12.0	1523—1518	Neuvonen and Grundström, 1969
9	Loftahammar gabbro, S	23.0 N	179.0	2.0	5.0	1694—1460	Poorter, 1976
10	Dala porphyries, S	23.0 N	184.0	—	—	1634	Mulder, 1971
11	Dala sandstone, S	32.6 N	176.5	6.9	13.5	Jotnian	Piper and Smith, 1980
12	Öje diabase, S	27.3 N	188.7	5.3	10.7	(935—750)	Piper and Smith, 1980
13	Nordingrå granite, S	32.5 N	149.1	7.7	14.3	(1565)	Piper, 1979a
14	Nordingrå gabbro-anorthosite, S	36.9 N	138.1	3.6	6.2	1552	Piper, 1979a
15	Ragunda gabbro-anorthosite, S	51.6 N	166.6	5.7	8.9	(1292)	Piper, 1979a
16	Gävle granite, S	35.0 N	165.0	7.3	13.8		Piper, 1979a
17	Rådmansö gabbro, S	36.0 N	201.0	4.3	8.5		Piper, 1980a
18	Uppsala metabasite, S	47.0 N	195.0	6.0	11.0		Piper, 1980a
19	Almunge district, S	34.0 N	188.0	7.0	13.0		Piper, 1980a
20	Hälleforsnäs "A", S	27.0 N	167.0	5.0	9.0	1518	Piper, 1980a
21:1	Hälleforsnäs "B", S	29.0 N	273.0	8.0	14.0	1518	Piper, 1980a
21:2	Hälleforsnäs High T., S	0.6 N	252.2	—	—	1518	Piper, 1980a
22	Bo diorite, S	59.0 N	188.0	6.0	10.0		Piper, 1980a
23	Bångfjället complex, S	23.8 N	196.5	—	—	1520	Hahlbrauck, 1982
24	Ragunda dolerites 1, S	14.5 N	194.5	8.1	15.3		Piper, 1979b
25	Ragunda dolerites 2, S	64.2 N	154.5	21.5	28.4		Piper, 1979b
EW 1	East—West dykes 1, S	27.7 N	196.8	4.1	8.2	1530	This paper
EW 2	East—West dykes 2, S	61.7 N	176.1	9.3	13.7	1530	This paper
TD	Tuna dykes, S	21.1 N	180.4	5.2	10.0	1371	This paper

Central Scandinavian Dolerite Group							
26	Gävle dolerite, S	8.0 N	150.0	2.0	4.0		Poorter, 1976
27	Gnarp dolerite, S	11.0 S	159.0	7.0	10.0		Poorter, 1976
28	Nordingrå dolerite, S	5.0 S	158.0	5.0	9.0	1245	Poorter, 1976
29	Nordingrå baked rocks, S	2.3 S	157.4	3.0	5.0		Piper, 1979a
30	Ulvö dolerites, S	1.0 N	150.0	5.0	9.0		Magnusson and Larsson, 1977
31	Ulvö dolerites, S	4.0 S	153.0	4.04	6.0		Larsson and Magnusson, 1976
32	Särna type dolerite, S	5.3 N	158.6	2.8	5.0	1247	Piper and Smith, 1980
33	Lybergsgnupen dolerite, S	0.6 N	165.7	—	—		Mulder, 1971
34	Åmån dolerite, S	27.1 N	163.0	—	—		Mulder, 1971
35	Satakunta sandstone, F	3.0 N	180.0	8.0	12.0	(1300)	Neuvonen, 1973
36	Satakunta dolerite, F	2.0 N	158.0	3.0	5.0	1236—1225, 1263	Neuvonen, 1965
37	Vaasa dolerite, F	7.0 N	164.0	3.0	6.0	1225, 1270	Neuvonen, 1966
38	Märket dolerite, F	5.9 S	145.5	7.0	11.0	1270	Neuvonen and Grundström, 1969
39	Vigehavn and Bölshavn A dolerites, Bornholm, D	14.0 N	155.8	7.3	13.5		Abrahamsen, 1977
45	Nordingrå sandstone, S	1.3 N	151.8	—	—		Magnusson, 1983
D 1	Emådalen dolerite, S	27.3 N	192.2	7.9	15.7	1223	This paper
D 2	Älvho dolerite, S	17.8 N	170.2	6.4	12.5	1215	This paper
D 3	Älvdalsåsen dolerite, S	1.4 S	148.6	3.7	6.2	1231	This paper
D 4	Mossi dolerite, S	7.8 N	151.3	20.0	37.5	1217	This paper
D 5	Bunkris dolerite, S	24.1 N	191.9	1.7	3.3	(1546—1516)	This paper
D 6	Glysjön dolerite, S	35.4 N	171.4	4.4	8.5		This paper
J 1	Sundsjö dolerite, S	11.0 S	161.5	15.7	22.9	1213—1156	This paper
J 2	Gimån dolerite, S	5.4 S	150.0	6.0	9.9	1229—1179	This paper
Sveconorwegian dolerites							
40	Bo dolerite, S	4.0 S	244.0	6.0	10.0		Piper, 1980a
41	Årby dolerite, S	7.3 S	227.4	6.8	9.5	985	Patchett and Bylund, 1977
42	Falu dolerite, S	6.1 S	237.6	4.6	7.1	966	Patchett and Bylund, 1977
43	Bräkne-Hoby dolerite, S	22.2 N	251.9	3.1	6.1	975	Patchett and Bylund, 1977
44	Tuve dolerite	17.0 S	239.0	2.0	3.0		Abrahamsen, 1974
SN 1	Mårsätter dolerite, S	31.6 S	258.5	18.0	19.5		This paper
D 9	Ejen dolerite, S	6.0 N	245.8	4.8	9.1		This paper

d_p and d_m are semiaxes in the 95% ellipses of confidence. The ages presented are either given in the palaeomagnetic reference or discussed in the text. Ages in parentheses are uncertain. D = Denmark, F = Finland, S = Sweden.

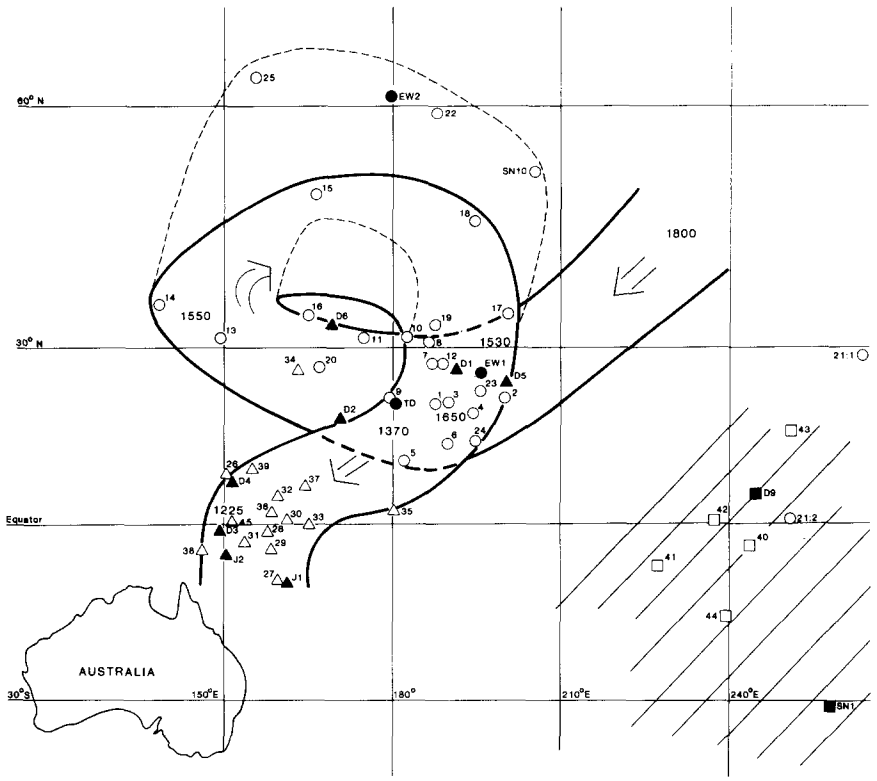


Fig. 12. A tentative APWP based on the data listed in Table II. The scratched area is the northern part of the Sveconorwegian loop (Bylund, 1981). Circles denote pre CSDG poles, triangles the CSDG poles and the squares Sveconorwegian poles. Filled symbols are for poles presented in this study. Either north or south pole is plotted.

two phases of dyke intrusions in the Ragunda area, with one, represented by pole 25, closely linked to the intrusion of the Ragunda anorthosite-gabbro and the other considerably younger.

However, the small number of poles, lack of radiometric age determinations and the dubious nature of the Group EW2 pole make the dashed part of the APWP provisional. The alternative is that the path trends eastwards and south from the Ragunda pole (15) and over poles 17, 18 and 19. They are all from metabasites in the area of Uppsala north of Stockholm (Piper, 1980a) and they may represent a late stage in the regional Svecokarelian cooling. In addition, pole 19 is from metabasites in the Almunge district where their NRM can have been influenced by the intrusion of alkaline rocks with a K—Ar biotite age of 1585 Ma (Doig, 1970).

The path then overlaps itself with a large number of poles in the overlapped area. The critical question then is which poles belong to the overlapped and which belong to the overlapping parts of the path. The Tuna dykes (TD) have an age of 1371 ± 50 Ma (Patchett, 1978) and are con-

sidered to have intruded at that time. Pole TD may then be used for fixing the overlapping part of the path while the previously discussed Finnish poles 3 to 6 represent the overlapped part.

For the E—W trending dykes Patchett (1978) reported whole-rock and mineral ages from the Eskilstuna dykes, sites SN12 and SN14, the western part of the Breven dyke, site SN3 and from rheomorphic minerals in Hällefors dyke, site SN10. The ages obtained range between 1560 to 1510 Ma and Patchett suggested that the dyke complex was emplaced at c. 1530 Ma. However, several of his measurements showed isotopic disturbances at later time in the Breven and Hällefors dykes, a tentative Sveconorwegian age for this disturbance was suggested. This is further enhanced by the presence of dykes of that age in the region of the E—W dykes, poles 40 and 41 in Table II and Fig. 12 and sites 41 and P59 in Fig. 1. Those disturbances can have affected the NRM of the E—W dykes and explain the large scatter in directions observed at some sites, e.g., SN5, SN8 and SN10 in Table I. This may also explain the differences between Piper's "A" direction, pole 20, of the Hällefors dyke (Piper, 1980a) and the results from the same part of the Hällefors dyke, site SN10, presented in this paper. In addition, Piper isolated a high-coercivity, high-temperature "B" direction (pole 21:1) and another high-temperature (c. 670°C) component (pole 21:2), both considered to be residing in haematite resulting from hydrothermal and deuteric activity (Piper, 1980a). Poles 21:1 and 21:2 are situated within the Sveconorwegian field in Fig. 12 and can be compared with the Sveconorwegian poles 41 and 43 in Fig. 12. This further indicates the presence of a Sveconorwegian component in the NRM of the Breven and Hällefors dykes.

Site SN1, Mårsätter, is situated in what is supposed to be an E—W trending dyke, the Viksjö dyke (Blomberg, 1901), but the magnetic direction gives a palaeopole closer to the Sveconorwegian area (Table II and Fig. 12). This opens the question if site SN1 represents a dyke of Sveconorwegian age or a remagnetized part of a dyke belonging to the E—W trending 1530 Ma dyke system.

The magnetic complexity displayed by the E—W dykes and the mix with younger dykes in the same area warrants another study based on a large sample collection representing all the various rock types that are present in the dykes. As for the present, the Group EW1 magnetic direction obtained by demagnetization at 100 mT or between 500 and 570°C is considered as representing the E—W trending dyke system at the time of its emplacement. This is supported by the agreement with the palaeopoles 7 and 8 of the Föglö-Sottunga dykes (Pesonen and Neuvonen, 1981) and the Föglö dykes (Neuvonen and Grundström, 1969), age 1523—1518 Ma; poles 1 and 2 from gabbros in northern Sweden supposed to have acquired their characteristic remanence at c. 1530 Ma, (Piper, 1980a; Elming, 1982, and pole 23 from the Bångfjället Window in the Scandinavian Caledonides for which Hahlbrauck (1982) presented a Rb—Sr age of 1520 ± 140 Ma.

Mulder (1971) reported K—Ar ages of 921—605 Ma for the Dala porphyries (10) while Welin and Lundqvist (1970) presented a Rb—Sr age of 1634 ± 38 Ma for the Dala porphyries in the Los-Hamra region c. 60 km E of Mulder's sites and this latter age is interpreted as the formation age of the porphyries. This places pole 10 in the overlapped part of the path.

The effusive Öje diabase (Jotnian lavas), pole 12, are interbedded in the Jotnian Dala sandstone (11). Its age is bracketed by the age of the Dala porphyries (1634 Ma) and the intrusion of the Dala complex of the CSDG at c. 1250 Ma. The fact that the Nordingrå granite, with a supposed age of c. 1550 Ma, is capped by the Jotnian Nordingrå sandstone, can restrict the period of Jotnian sedimentation in Scandinavia to between c. 1550—1250 Ma, and this gives a rough estimate of the Öje diabase. Priem et al. (1968) reported K—Ar whole-rock ages for the Öje diabase of between 935 and 750 Ma. The question is if this age and the similar K—Ar age obtained for the Dala porphyries are due to Ar-loss or to some geological event that reset the K—Ar system — and at the same time could have disturbed the original NRM of the rocks. A possible event could be uplift and cooling of the area, with the rocks passing the blocking temperature interval for the K—Ar system and the magnetic mineral assemblage present during the uplift. Nyström (1982) pointed to the presence of pumpellyite and actinolite in parts of the area covered by the sediments and diabases, and this would imply a depth of burial of several kilometres and a temperature of up to 300°C. If the rocks were buried for a prolonged period they may have obtained a postformation NRM which became blocked in when the rocks cooled due to uplift of the block. This process of remagnetization has been demonstrated from Greenland by Beckman and Mitchell (1976), Morgan (1976) and from southern Sweden by Bylund (1981). The theoretical considerations of the process have been treated by Pullaiah et al. (1975). However, such a remagnetization of the rocks in the northern part of the province Dalarna; porphyries, sandstones and diabases, must have occurred before the intrusion of the Dala complex of the CSDG at c. 1250 Ma in the same area as they show no signs of such a metamorphism (Nyström, 1982). Further, the palaeomagnetic directions in the CSDG-dolerites from the sandstone/porphyry area coincide with directions from the CSDG elsewhere, which further disproves the idea of a metamorphic event in northern Dalarna between 935 and 605 Ma as suggested by the K—Ar datings. In their study of the Dala sandstone and Öje diabase Piper and Smith (1980) presented examples of deviating patterns in their magnetic behaviour during demagnetization but conclude that their results, poles 11 and 12, represent the time of deposition and lithification of the sandstone and the time of deuteric-thermal alteration of the diabases.

The Loftahammar gabbro pole (9), is from a basic intrusion connected with metasediments and gneissic granites at Västervik in southeastern Sweden. Different dating methods have given the ages 1845 Ma (U—Pb concordia), 1620 ± 40 Ma (Rb—Sr whole-rock) and 1561 Ma (K—Ar reference line)

for the gneissic granites in the Loftahammar district (Åberg, 1978). Priem and Bakker (1973) presented an age of 1665 ± 35 Ma (Rb—Sr, whole-rock) for the gneissic granites. Priem et al. (1969) reported K—Ar biotite ages of 1465 ± 30 Ma for the granitic gneissic rocks in the Västervik area and Åberg (1978) presented a K—Ar whole-rock and mineral age of 1501 Ma for metasediments in the same area. Of the ages reported, the Rb—Sr ages may represent the time of the metamorphisis of the granite and the K—Ar ages later cooling. In such a case a correspondence between the K—Ar ages and the age of the magnetization of the gabbro is possible, i.e., that the Loftahammar gabbro pole (9) represents an age between 1560—1460 Ma.

The second Ragunda dolerite dyke pole (24) is from undated rocks, but the dolerite is younger than the Ragunda anorthosite-gabbro intrusion which supposedly occurred around 1550 Ma. The position of the pole suggests an age between 1530 Ma, the age of the Group EW1 pole and 1370 Ma, the time of the emplacement of the Tuna dykes.

Previously published palaeopoles from the Central Scandinavian Dolerite Group plot in a well-defined group at the equator between c. 145 and 165° E (Fig. 12). To this group connect poles D3, D4, J1 and J2 of this study. There is good age control for the main part of the CSDG with Rb—Sr ages ranging between 1290 and 1170 Ma for the intrusions studied (Table II) with a close agreement at c. 1220 Ma for whole-rock biotite pairs. This is in accordance with a K—Ar "isochrone" of 1245 Ma for dolerite sills of the Ulvö complex within the Nordingrå area (Welin and Lundqvist, 1975).

The good agreement between the K—Ar and Rb—Sr ages and the consistency in pole positions from different parts of the CSDG area imply that the characteristic magnetic remanence found in the CSDG dolerites was acquired at the time of their intrusion. But there are data from five sites in the central and eastern part of the Dala complex that present deviating magnetic directions. The sites are Nos. D1, D2, D5, D6 and 34 (Table II and Figs. 11 and 12). Sites D5 and D6 are the same as Mulder's (1971) sites ZDC and ZDE, and the data from Mulder are combined with those from this study in Table II and Fig. 12. Further sites D1 and 34 (Mulder's site ZDA) are from the same sill and sites D5 and D6 from the same dyke according to the map accompanying Hjelmqvist's (1966) description of the rocks of the province Dalarna (Kopparberg county). Both D1 and D2 are from monzonitic parts of the dolerites. Patchett (1978) gave ages of 1223 ± 36 Ma and 1215 ± 18 Ma for sites D1 and D2 and 1516 ± 62 and 1546 ± 84 Ma for site D5. The D1 and D2 ages are in accordance with other CSDG ages (Table II) but the D5 age is too old. For this site Patchett showed that clinopyroxene gives values below the feldspar isochron obtained and further that high initial $^{87}Sr/^{86}Sr$ ratios suggest substantial addition of radiogenetically enriched Sr to this dyke, a matter which could have produced an anomalously old age. Consequently Patchett gives little confidence to the ages obtained on rocks from site D5. The palaeopole for D5, on the

other hand, does agree well with poles of age c. 1550—1520 Ma, but so do poles D1 and D2 with good CSDG-ages (Fig. 12). Åberg and Bollmark (1983) reported a two-point whole-rock reference line of 1646 Ma and a whole-rock-mineral isochron age of 1515 ± 21 Ma (Rb—Sr) for clasts of a composite dyke from within the Siljan ring impact structure; c. 20 km SE of site ZDA in Fig. 12. They interpreted the 1646 Ma age as the emplacement age and the younger age as due to disturbances in the isotopic system caused by the meteorite impact at c. 360 Ma. This indicates the presence of more than one generation of dykes in the area: (1) a set of dykes with an age of about 1600—1500 Ma; and (2) the CSDG dykes. But this does not solve the problem with the conflicting radiometric ages and palaeopolpositions observed in the dolerites at sites D1 and D2. Tectonic movements within the area may be responsible for the discrepancies but no signs of such movements have been reported. A study is going on to see if the Siljan meteorite impact affected the NRM of the rocks within and just outside the impact area but it is not probable that the impact influenced rocks as far away from the impact site as e.g., sites D5 and D6.

The poles of the Dala sandstone (11) and the Satakunta sandstone (35) differ by about 30°. This could mean a real age difference, the Satakunta sandstones being deposited around 1300 Ma as suggested by Neuvonen (1973) and the Dala sandstones sampled by Piper and Smith (1980) being older. The Jotnian Nordingrå sandstone pole (45) is located within the cluster of CSDG poles proper and this is interpreted as due to a remagnetization of the sandstone by the capping dolerites of the Ulvö complex of the CSDG (Magnusson, 1983).

Pole 39 is a combined pole from two dolerite dykes on the Danish island of Bornholm (Fig. 1), the Vigehavn and Bölshavn A dykes. They are not dated but the palaeomagnetic results suggest that those dykes form a southern outlier to the CSDG (Abrahamsen, 1977).

Data from one dolerite dyke, D9 (Ejen), just south of the Dala complex of the CSDG (Fig. 7), presents a pole position in the area of the Sveconorwegian poles (Fig. 12). This implies that this dyke belongs to a set of dykes that follow the eastern margin of the Protogine Zone. This zone, also called the Sveconorwegian Front Zone or the Schistosity Zone, forms the border between the Sveconorwegian and Svecokarelian parts of Fennoscandia. Owing to stresses caused by uplift of the Sveconorwegian block, fissures were created through which the marginal dolerites intruded (Patchett and Bylund, 1977). As previously mentioned (p. 21) there is a possibility that the dolerite dyke represented by site SN1 also belongs to this set of marginal dykes. In Fig. 12 the poles of sites SN1 and D9 are plotted together with poles from the Sveconorwegian marginal dykes of Bo (40) and Årby (41), both situated among the E—W 1530 Ma dykes (Fig. 1), and the Falu dyke (42) in the province of Dalarna. The Årby dyke is dated at 995 ± 65 Ma and the Falu dyke at 966 ± 20 Ma (Patchett, 1978; Rb—Sr).

Another site, D7 (Fig. 12) is from a dolerite dyke in northern Dalarna.

It is the same as site 43 in Piper and Smith (1980). Patchett (1978) dated this dyke and assumes an age of c. 920 Ma as the time of its intrusion and that it belongs to the Sveconorwegian marginal dykes. Both palaeomagnetic studies present steep positive inclination and a combined pole at lat. 79° S and long. 168° E. This pole position does not agree with any published palaeopoles from Fennoscandia. The reason for this discrepancy is not known, but it can be noted that this site is not far from the CSDG sites D5 and D6 with their deviating magnetic directions. This may further imply that secondary events have influenced the NRM of the rocks in this area in northern Dalarna.

CONCLUSIONS

The palaeomagnetic data from the three sets of mafic rocks presented in this paper help to define the APWP for Fennoscandia between 1650 and 1200 Ma (Fig. 12). It is based on 51 independent palaeomagnetic poles, which is about one per 10 Ma. This figure can be compared with the North-American Precambrian average of 1.6 poles per 10 Ma (Roy, 1983). The Fennoscandian poles are clustered, 16 poles form the well-defined CSDG-cluster at the equator and there are 16 poles in the area where the path overlaps itself between 35—10° N and 180—200° E, and there are also five deviating CSDG poles. This leaves only 14 poles for the remaining part of the APWP. Of those the four Swedish anorthosite-gabbro and rapakivi granite poles are used for constructing the longitudinal loop at intermediate latitude and four other poles form the suggested high latitude hump.

The main feature of the APWP is of slow apparent polar wander, c. 40° in the period 1650—1200 Ma, 1° per 10 Ma, interrupted by a more rapid longitudinal excursion, the Sub-Jotnian loop, and a possible excursion into higher latitudes as suggested by the hump in Fig. 12. How this APWP reflects the movements of Fennoscandia during this period is difficult to discern as longitudinal movements are not possible to observe by the palaeomagnetic method. But calculations using the formula $\tan I = 2 \tan \lambda$ (I = inclination, λ = palaeolatitude) are possible. This formula, used on the Group EW1 data (1530 Ma) and the CSDG data (1225 Ma), indicate a possible latitudinal drift of 30° for central Fennoscandia during the actual time period, assuming that the equator was crossed. If not, the possible latitudinal drift is reduced to c. 18°. This corresponds to a drift of 1—0.5° per 10 Ma or roughly 1 cm a^{-1} in a north to south direction between c. 1530 and 1200 Ma.

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